Module 1: Introduction to AIOps & GitHub (8 Hours)

- Introduction to AIOps: Overview, Importance, and Benefits
- Core Concepts: Observability, Automation, and Intelligent Monitoring
- Understanding Git and GitHub Fundamentals
    - Repository creation, management, and branching strategies
    - Committing, pushing, and pulling code changes
- Integration of GitHub in IT Operations
- Collaborative Development and CI/CD practices
- Lab: Setting up GitHub Repository & Project Management Essentials

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Module 2: Data Management for AIOps using GitHub (8 Hours)

- Data Sources: Logs, Metrics, Events, and Traces
- Data Collection and Aggregation Methods
- Data Cleaning, Preparation, and Storage Techniques
- GitHub for Data Version Control and Management
    - Using GitHub Large File Storage (Git LFS)
    - Managing data pipelines and workflows in GitHub Actions
- Hands-on Lab: Data preprocessing and storing datasets on GitHub repositories

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Module 3: Machine Learning & Analytics in AIOps with GitHub (8 Hours)

- Machine Learning Foundations in AIOps
    - Anomaly Detection & Predictive Analytics
- Practical Algorithms: Classification, Regression, Clustering
- Deploying ML models with Version Control on GitHub
- Model Tracking, Experimentation, and Evaluation
- Leveraging GitHub Actions for Automated ML workflows
- Hands-on Lab: Developing, deploying, and tracking ML models using Python (Scikit-Learn, TensorFlow) integrated with GitHub Actions

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Module 4: Automation & Incident Response with AIOps and GitHub (8 Hours)

- Intelligent Event & Incident Management
    - Root Cause Analysis (RCA), Correlation, and Alerting
- Automating Response Workflows using GitHub Actions
- Integration of GitHub Actions with Incident Management Tools (Jira, ServiceNow, Slack)
- Automated testing and validations for infrastructure and code deployments
- Monitoring, logging, and feedback mechanisms
- Hands-on Lab: Automating alerts, remediation, and CI/CD pipelines using GitHub Actions workflows

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Module 5: Advanced AIOps Deployment & Best Practices with GitHub (8 Hours)

- Designing scalable AIOps Architectures
- Cloud-based Deployment vs. On-premises Deployment
- Security, Compliance, and Governance in GitHub repositories
- Managing Infrastructure as Code (IaC) with GitHub
- Real-world Case Studies: GitHub in enterprise-scale AIOps solutions
- Continuous Integration/Continuous Deployment (CI/CD) Best Practices
- Capstone Project: End-to-end AIOps Implementation using GitHub
    - Design, Deploy, Manage, and Automate an AIOps solution with GitHub integration